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Handwritten Notes & Further Reading — Prof. Fei-Fei Li Distinguished Lecture

Talk title	What we see and what we value: AI with a human perspective
Speaker	Prof. Fei-Fei Li (Stanford University)
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Talk URL/link	https://www.youtube.com/watch?v=gzOwpEupP5w
Talk was a	Distinguished Lecture
Author	Sanjeev Hiremath

Handwritten notes by Sanjeev, cleaned up using Claude Cowork.

Abstract

Vision emerged in prehistoric animals more than 540 million years ago, and visual intelligence has been a cornerstone of animal intelligence ever since — so enabling machines to see is a critical step toward building intelligent machines. The talk walks through three arcs: computers seeing what humans see (deep learning and ImageNet), computers seeing what humans *don't* see (ambient intelligence in healthcare; privacy and bias in vision), and computers seeing what humans *want* machines' help seeing and doing (smart cameras, household robots).

The cumulative lesson — AI developed from a human-centered perspective — is the founding premise of Stanford's Institute for Human-Centered AI (HAI).

Bio

Dr. Fei-Fei Li is the inaugural Sequoia Professor in Stanford's Computer Science Department and Co-Director of Stanford's Human-Centered AI Institute; she previously directed the Stanford AI Lab (2013–2018) and served as Chief Scientist of AI/ML at Google Cloud during a sabbatical (2017–2018). She is the inventor of **ImageNet** and the ImageNet Challenge, an elected member of the NAE, NAM, and AAAS, and a Fellow of ACM. She is also the author of *The Worlds I See: Curiosity, Exploration and Discovery at the Dawn of AI* (Macmillan, 2023).

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1. Framing — Vision as the Big Bang of Evolution

First Light — ~540 million years ago. From Earth's primordial soup, over a geologically short period (~10 million years), there was an explosion of animal species — the **Cambrian Explosion**, sometimes called the "Big Bang of evolution."

Many theories explain it — climate change, chemical composition of water, etc. One leading theory, from Australian zoologist **Andrew Parker**, conjectures that the speciation explosion was triggered by the sudden evolution of **vision**. Light became information. Once animals could see, selection pressure on predator/prey behavior, navigation, and mating exploded.

Prof. Li's framing: vision isn't decoration — it is arguably what kicked off intelligence on this planet. That sets up the rest of the talk.

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2. Three Theses of the Talk

Where have we come from, and where are we heading?

1. **Building AI to see what humans see.**
2. **Building AI to see what humans *don't* see.**
3. **Building AI to see what humans *want* to see.**

These three threads structure the rest of the hour.

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3. Vision and Computer Vision

- Humans are **effortless, accurate, and fast** at seeing: object categorization happens in under **150 ms**.
- We have **dedicated neural correlates** in the brain — areas that specialize in visual recognition.
- **Object recognition is a fundamental building block** of visual intelligence — a north-star problem.
- Vision lets us navigate, communicate, entertain, socialize. "You see the world in a fundamentally different way."

Computer vision is ~60 years old. It was famously estimated as a summer project at MIT (July 7, 1966 — Seymour Papert's "The Summer Vision Project"). It didn't finish that summer. It has since become a thriving field — both academically and economically — powering image classification, autonomous driving, DALL·E-style generation, and more.

Why is it hard? A 3D world projects onto 2D pixels in mathematically infinite ways. Add differences in:

- Lighting or texture
- Background or occlusion
- Viewing angles

…and you have a *really* hard problem.

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4. Three Phases of Object Recognition

Phase 1 — Hand-designed features *and* hand-designed models

- **Geons → Recognition by Components.** Biederman's theory that any visual object decomposes into ~24 primitive 3D shapes (cylinders, blocks, wedges, …) arranged in relation.
- **Generalized Cylinders.** Binford (1971) — model each object as a cylinder whose cross-section can vary in shape/size along a curving axis.
- **Parts + Springs.** Fischler & Elschlager (1973) — represent an object as part templates connected by deformable "springs" encoding spatial relations.

"Theoretically beautiful, but they don't work."

Phase 2 — Hand-designed features, learned models

- **Pictorial Structures & Constellation Models.** Probabilistic descendants of Parts + Springs — detect parts independently, score their spatial arrangement under a prior.
- **Non-parametric Bayes.** Let the number of parts / categories / clusters grow with the data instead of fixing it up front (Dirichlet-process style).
- **Boosting.** Combine many weak classifiers into a strong one; Viola–Jones face detection is the canonical win.
- **Conditional Random Fields.** Graphical model over labels given features — captures neighbor dependencies; strong for segmentation / structured output.
- **Bag of Words.** Borrow from NLP: represent an image as an unordered histogram of "visual words" (SIFT descriptors quantized into a vocabulary).
- **And-Or Graphs.** Compositional grammar — "And" nodes decompose into parts, "Or" nodes enumerate alternatives; explore multiple structural hypotheses.
- Datasets of the era: **Pascal VOC**, **Caltech 101**
- Prof. **Biederman**'s conjecture: humans know **~30K–100K** visual categories

Phase 3 — Learned features *and* learned models (2012 → present)

- **ImageNet** — 15M images, ~22K categories.
- Brought back **Convolutional Neural Networks** — a classic algorithm.
- **2012** → Pre-Transformer CNN era begins.

Slide: "ImageNet Challenge — Classifying 1000 Object Classes."

- 1000 object classes, 1,431,167 images.
- Top-5 classification error by year:
  - 2010 — 28% (shallow models)
  - 2011 — 26% (shallow models)
  - 2012 — 16% → AlexNet
  - 2013 — 12%
  - 2014 — 7%

  - 2015 — 3.6% (ResNet — human baseline ~5% crossed)
  - 2016 — 3%
  - 2017 — 2.3%
- Refs: Deng, Dong, Socher, Li, Li, Fei-Fei CVPR 2009; Russakovsky et al. 2015.

Slide: "ImageNet Enabled Tremendous Progress in Computer Vision" (model timeline, Wu/Sahoo/Hoi *Neurocomputing* 2020).

- 2012 — AlexNet
- 2013 — OverFeat; R-CNN
- 2014 — VGG Net; GoogLeNet; Fast R-CNN
- 2015 — ResNet; Faster R-CNN
- 2016 — FPN; YOLO; SSD; DenseNet; ResNet v2
- 2017 — R-FCN; YOLO9000; ResNeXt; Hourglass; DPN; MobileNet; RetinaNet
- 2018 — DCN; RefineDet; Mask R-CNN; SENet; NASNet
- 2019 — Cascade R-CNN; FSAF; CornerNet; ExtremeNet; CenterNet; FCOS; NAS-FPN; Detnas; EfficientNet

Landmark steps: **AlexNet** (first deep CNN to win ImageNet), **VGG** (simple deep 3×3 stacks), **GoogLeNet** (Inception modules), **ResNet** (skip connections unlock 100+-layer nets, crosses human baseline), **Mask R-CNN** (instance segmentation), **EfficientNet** (compound scaling).

Vision contributed to deep learning — not just benefited from it. ImageNet was one of the catalysts of the whole deep-learning revolution.

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5. Beyond Objects — Scene Graphs & Compositionality

Objects alone aren't enough to categorize a scene — **relationships between objects must also be modeled**.

- **Scene Graph Representation** of visual relationships: entities, attributes, inter-entity relations.
- **Visual Genome** (Prof. Li's project) operationalizes this.
- **Compositionality → long-tail relations.** Representing scenes combinatorially lets models generalize to rare/unseen combinations:
  - *"person riding horse"* + *"person wearing hat"* → zero-shot recognition of *"horse wearing a hat."*
  - *"person sitting on chair"* + *"fire hydrant"* → zero-shot recognition of *"person sitting on fire hydrant."*

Compositionality at inference time is how you get coverage over the combinatorial long tail without needing a labeled example of every combination.

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6. Static → Dynamic Relations

From static scenes to activities over time.

- **MOMA — Multi-Object Multi-Actor Activity understanding.** Example: *"couch + chair + players playing ping pong."*
- Related threads: 3D vision, DALL·E-style generation, image segmentation, pulse segmentation.

Observation: Data + compute + neural nets/algorithms ushered in the deep-learning revolution in AI. AI's development has been — and will continue to be — inspired by brain science and human cognition.

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7. Thesis 2 — Building AI to See What Humans *Don't* See

Humans are bad at certain visual tasks. AI can fill those gaps.

Fine-grained recognition

- Species of birds.
- Car make / model / year (fine-grained "brand detectors").
- Street View + 2010 Census correlation → the world through a vision lens infers **income and voting patterns** from neighborhood imagery.
- Computers can pass a **Stroop-test-style** setup that trips humans up.

Medical settings — things humans miss

- Missing surgical instruments after an operation.
- **Priming bias** — the effect where what you expect to see makes you miss what's actually there (gorilla-in-the-lung-scan type stuff).
- **Hospital-acquired infections** kill roughly **three times more people** in the US than car accidents every year.
- Depth sensors measuring **hand hygiene** compliance.
- Patient-mobility detection & tracking.

→ Ambient Intelligence for healthcare. Smart depth sensors illuminating "dark spaces" in hospitals. A consistent framing: the goal isn't surveillance, it's filling blind spots where human attention demonstrably fails.

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8. Thesis 3 — Building AI to See What Humans *Want* to See

AI triggers anxiety (labor threat) — but the US is also missing ~1 million nurses

AI here is framed as **augmenting human capabilities**, not replacing them. Medical examples:

- We don't know how the patient is doing.
- We don't know care quality.
- We don't know where the small instrument is.
- We don't know pharmacy errors.
- → Ambient Intelligence for healthcare (again).

Embodied AI — firefighters, doctors, everyday households *(screenshots)*

Slide: "An Opportunity — Computer vision and NLP progress inspires a *new north star* for robotic learning and embodied AI."

- Computer vision — ImageNet, 15M.
- NLP — SQuAD, Common Crawl.
- "Large data drives learning so much."

Slide: "Most robotics research today — Artificially simple environments." Three failure modes:

- × Small scale & anecdotal.
- × Experimenter-picked tasks & setup.

- × Lack of standard benchmark & metrics.

Slide: "Most robotics research today — Skill-level tasks, short-horizon goals, closed-world instructions." Classical methods shown (same Phase-2 vocabulary as object recognition — the robotics side is behind by an era):

- Pictorial Structure & Constellation Models
- Boosting
- Bag of Words
- Non-parametric Bayes
- Conditional Random Field
- And-Or Graphs

Slide: "A new north star — An ecological robotic learning environment + a large and diverse set of activities."

- **BEHAVIOR** — **B**enchmark for **E**veryday **H**ousehold **A**ctivities in **V**irtual, **I**nteractive, **a**nd **E**cological envi**R**onments.

Slide: "Focusing on people — 1000 household activities that people prefer more help with."

- Activity rank 1–2000; preference score curve — first 1000 selected.
- Originated from human needs (serves human needs, goals, values).
- Sample activities:
    1. Clean after wild party
    2. Clean a shower
    3. Clean grease
    4. Scrub bathroom floor
    5. Shovel snow
    6. Wash outside window
    7. Clean walls
    8. Clean up water damage
    - …
    1983. Open b-day presents
    1984. Open Christmas gifts
    1985. Play snooker
    1986. Throw darts
    1987. Buy a strapless bra
    1988. Mix baby cereal
    1989. Buy a ring
    1990. Play squash

Slide: "A focus on ecological and diverse scenes and objects."

- 1200+ categories.
- 5000+ 3D models.
- 30+ properties. Sample properties: Assembleable, Breakable, Boilable, coldSource, Fillable, Freezable, Foldable, Heatable, heatSource, Sliceable, Soakable, softBody, Stickable, Toggleable, Washable, waterSource.
- Compared to similar work from UW (Objaverse).

Slide: "Simulation environments are critical for robotic learning."

1. Fast training, transferable with sim2real.
2. Safe for robots and environment.
3. Towards reproducible benchmark for interactive tasks.

- iGibson / ImageNet-referencing lineage.

Slide: "BEHAVIOR's new simulation environment — OmniGibson aims at realism in *physics*, perception & interactions." (NVIDIA Omniverse collaboration.)

- Thermal effects.
- Transition machine (phase change).
- Lights and reflections.
- Fluids.
- Deformables.
- Transparency.

Structure in motion planning and action *(screenshots)*

Slide: "What structure can we introduce to the problem?" (VoxPoser, CoRL 2023 — Huang, Wang, Zhang, Li, Wu, Fei-Fei.)

- Instruction: *"Open the drawer."*
- Pipeline: **LLM + VLM → Code → Code-Generated 3D Value Maps → Motion Planning → Actions.**

Slide: VoxPoser instruction example — "Open the top drawer."

- VLM = Visual Language Model (grounds code to the real scene).
- LLM output (code):

```
def affordance_map():
    mask = np.zeros((100, 100, 100))
    map  = np.zeros((100, 100, 100))
    handles = detect('handle')
```

- Pattern: language model emits code; VLM resolves `detect('handle')` against the actual point cloud.

Privacy — "seeing too much"

- **PrivHAR** — Recognizing Human Action from a privacy-preserving lens. Warped-signal approach: humans aren't visible in the feed, but activity can still be recovered (hardware + software co-design).

Observation: AI can **amplify or exacerbate** many pre-existing issues (bias, privacy, etc.) that have plagued humanity for ages. We must commit to **study, forecast, and guide** AI's impact on people and society, in a **multi-disciplinary** way.

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9. What Do We Value? — Human-Centered AI (HAI)

Three letters, three commitments:

- **H — Human impact.** The development of AI must be guided by a concern for its human impact.
- **A — Augment, don't replace.** AI should strive to augment and enhance humans, not replace them.
- **I — Inspired by human Intelligence.** AI technologies can draw inspiration from human intelligence (brain science, cognition, development).

That framing — **HAI** — is Stanford's Human-Centered AI Institute mission in miniature, and it's the through-line of the whole talk.

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10. Closing Observations

- AI's development was ushered in by **data + compute + neural nets/algorithms** — and will continue to be inspired by **brain science and human cognition**.
- AI must **augment, not replace**, our individual and collective human capabilities and wellbeing.
- AI can amplify existing societal issues; a multi-disciplinary approach to studying, forecasting, and guiding its impact is required.

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11. Further Reading

- **Prof. Fei-Fei Li, *The Worlds I See* (2023)** — memoir + intellectual history of computer vision, ImageNet, and the founding of Stanford HAI. The most natural follow-up to the talk.
- **Stanford HAI (Human-Centered AI Institute)** — https://hai.stanford.edu/. Prof. Li co-founded this; it's the institutional home of the "HAI" framing.
- **ImageNet paper** — Deng, Dong, Socher, Li, Li, Fei-Fei, *"ImageNet: A Large-Scale Hierarchical Image Database,"* CVPR 2009. https://www.image-net.org/.
- **ImageNet Large Scale Visual Recognition Challenge (ILSVRC)** — Russakovsky et al., IJCV 2015. arXiv: https://arxiv.org/abs/1409.0575.
- **AlexNet** — Krizhevsky, Sutskever, Hinton, *"ImageNet Classification with Deep Convolutional Neural Networks,"* NeurIPS 2012. https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html.
- **Visual Genome** — Krishna et al., *"Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations,"* IJCV 2017. arXiv: https://arxiv.org/abs/1602.07332.
- **Scene Graph Generation (Xu, Zhu, Choy, Fei-Fei, CVPR 2017)** — arXiv: https://arxiv.org/abs/1701.02426. The message-passing scene-graph paper that underlies most of the compositionality work.
- **MOMA** — Luo, Fei-Fei et al., *"MOMA: Multi-Object Multi-Actor Activity Parsing,"* NeurIPS 2021. arXiv: https://arxiv.org/abs/2112.04886.
- **Ambient Intelligence in healthcare** — Haque, Milstein, Fei-Fei, *"Illuminating the dark spaces of healthcare with ambient intelligence,"* Nature 2020. https://www.nature.com/articles/s41586-020-2669-y.
- **Andrew Parker, *In the Blink of an Eye* (2003)** — the "Light Switch Theory" of the Cambrian Explosion that Prof. Li cites.
- **BEHAVIOR-1K** — Li, *Zhang*, Wong, *Gokmen*, Srivastava, *Martin-Martin*, Wang, *Levin*, Lingelbach, …, Fei-Fei, *"BEHAVIOR-1K: A Benchmark for Embodied AI with 1,000 Everyday Activities and Realistic Simulation,"* CoRL 2022. https://behavior.stanford.edu/behavior-1k.
- **OmniGibson / iGibson 2.0** — the simulation environment underlying BEHAVIOR (NVIDIA Omniverse collaboration). https://behavior.stanford.edu/omnigibson/.
- **VoxPoser** — Huang, Wang, Zhang, Li, Wu, Fei-Fei, *"VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models,"* CoRL 2023. arXiv: https://arxiv.org/abs/2307.05973.
- **Objaverse** (UW-adjacent contrast point from the talk) — Deitke, Schwenk, et al., *"Objaverse: A Universe of Annotated 3D Objects,"* CVPR 2023. arXiv: https://arxiv.org/abs/2212.08051.
- **PrivHAR** — Hinojosa, Niebles, Arguello, *"Learning Privacy-preserving Optics for Human Pose Estimation"* / PrivHAR line of work. arXiv: https://arxiv.org/abs/2208.10544.
- **Street View → demographics** — Gebru, Krause, Wang, Chen, Deng, Aiden, Fei-Fei, *"Using deep learning and Google Street View to estimate the demographic makeup of neighborhoods across the United States,"* PNAS 2017. https://www.pnas.org/doi/10.10

73/pnas.1700035114.